DAVID (YOON SUK) KANG

CONTACT INFORMATION

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RESEARCH INTERESTS

My primary research interests lie in data mining applied to diverse graph data types (*e.g.*, *conventional*, *signed*, *knowledge*, and *hypergraph* structures), with a particular emphasis on uncovering knowledge from real-world networks.

- **Conventional Graph**: Community detection (CIKM'20, KBS'22)
- **Signed Graph**: Community detection (WWW'26, ICDM'21, TKDE'23); Representation learning (TKDD'24)
- **Hypergraph**: Representation learning (KDD'26, IEEE Access'25); Hypergraph analysis (WWW'24)

(°: *Under Review*)

EDUCATION

Hanyang University, Seoul, Korea — Mar. 2013 – Feb. 2022
- *Ph.D. in Computer Science*
  - Thesis: Graph Reinforcement for Accurate Community Detection and Embedding on Graphs and Hypergraphs
  - Advisor: Prof. Sang-Wook Kim

Hanyang University, Seoul, Korea — Mar. 2007 – Feb. 2013
- *B.S. in Computer Science*

POSITIONS

Chungbuk National University, Cheongju-si, Chungcheongbuk-do, Korea — Sep. 2024 – Present
- *Assistant Professor, School of Computer Science*

RESEARCH EXPERIENCES

University of Michigan, Ann Arbor, MI, USA — May 2022 – Aug. 2024
- *Postdoctoral Researcher, School of Information*
  - Topic: Data Mining on Large-Scale Hypergraph
  - Advisor: Prof. Qiaozhu Mei

The Pennsylvania State University, University Park, PA, USA — Oct. 2019 – Feb. 2020
- *Visiting Scholar, College of Information Sciences and Technology*
  - Topic: Improving the Accuracy of Community Detection
  - Advisor: Prof. Dongwon Lee

AWARDS & HONORS

Received the **Young Researcher Award** — 2025
- Korean Institute of Intelligent Systems

Received the **Young Researcher Award** — 2025
- KIIS International Symposium on Advanced Intelligent Systems

Received the **Best Paper Award in Samsung Research Project** — 2022
- Samsung Electronics Co., Ltd.

Received the **Outstanding Ph.D. Dissertation Award** — 2022
- Research Institute of Industrial Science, Hanyang University

Awarded the **NAVER Ph.D. Fellowship** — 2021
- Naver Corporation

Received the **Best Paper Award** — 2021
- KIPS Spring Conference

Received the **ACM SIGIR Student Travel Award** — 2017, 2020
- ACM International Conference on Information and Knowledge Management (ACM CIKM)

Received the **ACM SIGAPP Student Travel Award** — 2016
- ACM Symposium on Applied Computing (ACM SAC)

Awarded the **NHN&HYU Ph.D. Fellowship** 2015

- NHN Corporation

Received the **Best Paper Award** 2014

- IEEE International Conference on Network Infrastructure and Digital Content (IEEE IC-NIDC)

PUBLICATIONS

Preprinted/Submitted Papers (* indicates equal contributions)

[1] Multimodal Recommender Systems
David Y. Kang
Under Review at One of Top-Tier DM Conferences

International Conference and Journal Papers

[16] HyperGC: Learning Hypergraph Representations via Full Hyperedge Reconstruction and Contrastive Learning
David Y. Kang*, So-Bin Jung*, and Sang-Wook Kim
KDD 2026 (*ACM International Conference on Knowledge Discovery and Data Mining*)
Full Paper (Acceptance Rate ≈ 18.5%)

[15] Improving the Accuracy of Community Detection on Signed Networks via Community Refinement and Contrastive Learning
Hyunuk Shin*, Hojin Kim*, Chanyoung Lee*, Yeon-Chang Lee, and David Y. Kang
WWW 2026 (*ACM Web Conference*)
Short Paper (Acceptance Rate ≈ 23%)

[14] Revisiting Clique and Star Expansions in Hypergraph Representation Learning: Observations, Problems, and Solutions
David Y. Kang, Eujeanne Kim, Kyungsik Han, and Sang-Wook Kim
IEEE Access (SCIE Journal, 2026)

[13] Trustworthiness-Driven Graph Convolutional Networks for Signed Network Embedding
Min-Jeong Kim*, Yeon-Chang Lee*, David Y. Kang, and Sang-Wook Kim
ACM Transactions on Knowledge Discovery from Data (SCIE Journal, 2024)

[12] Low Mileage, High Fidelity: Evaluating Hypergraph Expansion Methods by Quantifying the Information Loss
David Y. Kang, Qiaozhu Mei, and Sang-Wook Kim
WWW 2024 (*ACM Web Conference*)
Full Paper (Acceptance Rate ≈ 20%)
Selected for Oral Presentation

[11] A Framework for Accurate Community Detection on Signed Networks Using Adversarial Learning
David Y. Kang, Woncheol Lee, Yeon-Chang Lee, Kyungsik Han, and Sang-Wook Kim
IEEE Transactions on Knowledge and Data Engineering (Top 5% SCIE Journal, 2023)

[10] Community Reinforcement: An Effective and Efficient Preprocessing Method for Accurate Community Detection
Yoonsuk Kang, Jun-Seok Lee, Won-Yong Shin, and Sang-Wook Kim
Knowledge-Based Systems (Top 10% SCIE Journal, 2022)

[9] Adversarial Learning of Balanced Triangles for Accurate Community Detection on Signed Networks
Yoonsuk Kang*, Woncheol Lee*, Yeon-Chang Lee, Kyungsik Han, and Sang-Wook Kim
ICDM 2021 (*IEEE International Conference on Data Mining*)
Short Paper (Acceptance Rate ≈ 20%)

[8] FORESEE: An Effective and Efficient Framework for Estimating the Execution Times of IO Traces on the SSD
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Wan D. Bae, Wonjun Lee, and Sang-Wook Kim
IEEE Transactions on Computers (SCIE Journal, 2021)

[7] CR-Graph: Community Reinforcement for Accurate Community Detection
Yoonsuk Kang, Jun-Seok Lee, Won-Yong Shin, and Sang-Wook Kim
CIKM 2020 (*ACM International Conference on Information and Knowledge Management*)
Short Paper (Acceptance Rate $\approx$ 25%)

[6] A Framework for Estimating Execution Times of IO Traces on SSDs
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Wan. D. Bae, and Sang-Wook Kim
CIKM 2017 (*ACM International Conference on Information and Knowledge Management*)
Short Paper (Acceptance Rate $\approx$ 28%)

[5] The uFLIP Benchmark Revisited for Evaluating SSDs
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Sang-Wook Kim, and Young Kyun Shin
International Journal of Communication Systems (SCIE Journal, 2016)

[4] A Methodology for Estimating Execution Times of IO Traces in SSDs 4664
Yoonsuk Kang
SAC 2016 (*ACM Symposium on Applied Computing*)

[3] Exploiting the uFLIP Benchmark for Analyzing SSDs Performance
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Sang-Wook Kim, and Young Kyun Shin
IC-NIDC 2014 (*IEEE International Conference on Network Infrastructure and Digital Content*)
Received the Best Paper Award

[2] Running Data Mining Algorithms on SSDs
Yoonsuk Kang, Yong-Yeon Jo, Duck-Ho Bae, and Sang-Wook Kim
EDB 2013 (*International Conference on Emerging Databases-Technologies, Applications, and Theory*)

[1] Selecting Similar Users in Collaborative Filtering
Sang-Chul Lee, Yoonsuk Kang, Seihyun Jeong, Min-Hee Jang, Young-Sup Hwang, and Sang-Wook Kim
ICGHIT 2013 (*International Conference on Green and Human Information Technology*)

Domestic Conference and Journal Papers

[15] Mobile and VR Technologies for Social Independence in Autistic Individuals: A Scoping Review
Bogoan Kim and David Y. Kang
Journal of Korean Institute of Intelligent Systems (KCI Journal, 2026)

[14] Exploring the Potential of LLM as an Enhancer for Hypergraph Learning
JungHyun Kim, David Y. Kang, and Sang-Wook Kim
KSC 2025

[13] Constructing a Graph-Based arXiv Dataset By Reflecting the Research Trend in Computer Science
Juhyun Jeon, David Y. Kang, and Sang-Wook Kim
ASK 2024

[12] Evaluating the Performance of Hypergraph Embedding Methods According to Hypergraph Sparsity
So-Bin Jung, David Y. Kang, and Sang-Wook Kim
ASK 2024

[11] CoAID+: COVID-19 News Cascade Dataset for Social Context Based Fake News Detection
Soeun Han, Yoonsuk Kang, Yunyong Ko, Jiwon Ahn, Yusim Kim, Seongsu Oh, Heejin Park, and Sang-Wook Kim
KIPS Transactions on Software and Data Engineering (KCI Journal, 2022)

[10] COVID-19 Cascade Dataset for Fake News Detection
Soeun Han, Yoonsuk Kang, Yunyong Ko, Jiwon Ahn, Yusim Kim, Seongsu Oh, Heejin Park, and Sang-Wook Kim
KIPS Spring Conference 2021

Received the Best Paper Award

[9] A Preprocessing Method for Accurate Link Prediction on Social Networks
Seungbeom Son, Yeonsuk Choi, Yoonsuk Kang, and Sang-Wook Kim
KIPS Fall Conference 2020

[8] Performance Comparison of Similarity-Based Link Prediction in Social Networks
Jun-Seok Lee, Yoonsuk Kang, and Sang-Wook Kim
KCC 2019

[7] Performance Comparison of Community Detection Algorithms in Social Networks
Jun-Seok Lee, Yoonsuk Kang, and Sang-Wook Kim
KCC 2018

[6] A Method for Analyzing Features that Affect the Performance of SSD
Yoonsuk Kang, Yong-Yeon Jo, and Sang-Wook Kim
KIPS Spring Conference 2018

[5] Community Detection by Sub-Community and CScan
Chunghyeon Cho, Gunjoo Ahn, Yoonsuk Kang, Jiwon Hong, and Sang-Wook Kim
KDBC 2018

[4] A Methodology for Estimating Execution Times of IO Traces on SSDs
Yoonsuk Kang, Yong-Yeon Jo, Jaehyuk Cha, Wan D. Bae, and Sang-Wook Kim
KCC 2017

[3] Anaylzing the Performance of SSDs in OLTP Environment
Seoung-Hun Jeong, Jae-Sung Lee, Yoonsuk Kang, Yong-Yeon Jo, Duck-Ho Bae, Sang-Wook Kim, Juyoung Kang, and Jahyuk Cha
KIISE Fall Conference 2013

[2] Analysis on I/O Trace Replayer for SSD Performance Evaluation
Inhyuk Yee, Kyuhwan Lee, Yoonsuk Kang, Yong-Yeon Jo, and Sang-Wook Kim
KIPS Fall Conference 2013

[1] A Method for Selecting Similar Users for Collaborative Filtering
Yoonsuk Kang, Seihyun Jeong, Sang-Chul Lee, Min-Hee Jang, Sang-Wook Kim
KIPS Fall Conference 2012

INVITED TALKS

Finding Communities in Social Networks: Challenges and Solutions
- Invited Talk @Hanyang Univ., Apr. 2026

Self-Supervised Hypergraph Representation Learning
- Invited Talk @Hanyang Univ., Jun. 2025

Evaluating Hypergraph Expansion Methods by Quantifying the Information Loss
- Invited Talk @UNIST, Dec. 2025
- Invited Talk @Int'l Symp. on Advanced Intelligent Systems, Nov. 2025
- Invited Talk @Chung-Ang Univ., Nov. 2024

Knowledge Discovery from Real-world Relationships
- Invited Talk @CBNU College of Veterinary Medicine, Apr. 2026
- Invited Talk @Chung-Ang Univ., Sep. 2025
- Invited Talk @Hanyang Univ., Aug. 2024

Adversarial Learning of Balanced Triangles for Accurate Community Detection on Signed Networks
- Invited Talk @ METU-HYU Joint Workshop, Dec. 2022

FORESEE: An Effective and Efficient Framework for Estimating the Execution Times of IO Traces on SSDs
- Invited Talk @ Waseda-UMS-Hanyang-UKM (WUHU) Joint Workshop, Dec. 2017

TEACHING

Graduate School
- Graph Machine Learning & Social Network Analysis (8836665) — Spring 2026

Undergraduate School
- Data Science (5119010) — Spring 2025 –2026
- Data Structure (5118006) — Spring 2025 –2026
- Algorithms (5118013) — Fall 2024 – 2025
- Fundamental Computer Programming (0914002) — Fall 2024 – 2025
- Information Retrieval (5119006) — Fall 2025
- Sever Programming (5120017) — Fall 2024 – 2025
- Machine Learning (5119008) — Spring 2025

PROFESSIONAL SERVICES

Program Committee Member (or Conference Reviewer)
- ACM Conference on Research and Development in Information Retrieval (**SIGIR**) — 2024, 2026
- ACM Web Conference (**WWW**) — 2023, 2024, 2026
- ACM International Conference on Information and Knowledge Management (**CIKM**) — 2019, 2020, 2025, 2026
- ACM Conference on Knowledge Discovery and Data Mining (**KDD**) — 2021 – 2024
- IEEE International Conference on Data Mining (**ICDM**) — 2022 – 2024
- ACM Symposium on Applied Computing (**SAC**) — 2023 – 2026
- International AAAI Conference on Web and Social Media (**ICWSM**) — 2017
- International Conference on Database Systems for Advanced Applications (**DASFAA**) — 2016

Journal Reviewer
- Journal of Computing Science and Engineering (SCOPUS) — 2025
- Journal of Supercomputing (SCIE) — 2023

PATENTS

Granted Patents
- Hypergraph Embedding Method and Systems Based on Graph Convolutional Networks Considering Relationships of Multiple-users
  Registration Number: **KR10-2928208** — Feb. 2026
- Adversarial Leyarning of Balanced Triangles for Accurate Community Detection on Signed Networks
  Registration Number: **KR10-2764877** — Feb. 2025
- Method for Reconfiguration of a Community in a Network Including a Plurality of Networks and an Electronic Device for the Method
  Registration Number: **KR10-2409160** — Jun. 2022
- A Feature Extraction Apparatus and Method for Predicting the Execution time of the Query Input and Output Trace
  Registration Number: **KR10-2249832** — May 2021
- A SSD Performance Evaluation Apparatus and Method for Predicting the Execution time of the Query Input and Output Trace
  Registration Number: **KR10-1950801** — Feb. 2019
- Method for Selecting Similar Users for Collaborative Filtering Based on Earth Mover's Distance
  Registration Number: **KR10-1620659** — May 2016

Filed Patents
- Method and Apparatus for Improving Community Detection Accuracy Based on Community Refinement and Self-Supervised Learning

Application Number: **KR10-2026-0023858** Feb. 2026

- Method and System for Measuring the Amount of Information Loss of a Graph Obtained Through a Hypergraph Expansion Method
  Application Number: **KR10-2023-0109155** Aug. 2023

References

Qiaozhu Mei, *Professor* (Postdoc. Advisor) qmei@umich.edu
School of Information, University of Michigan

Kyungsik Han, *Associate Professor* (Collaborator) kyungsikhan@hanyang.ac.kr
Department of Data Science, Hanyang University

Dongwon Lee, *Professor* (Visiting Scholar Advisor) dongwon@psu.edu
College of Information Sciences and Technology, The Pennsylvania State University

Wan D. Bae, *Professor* (Collaborator) baew@seattleu.edu
Department of Computer Science, Seattle University

Sang-Wook Kim, *Professor* (Ph.D. Advisor) wook@hanyang.ac.kr
Department of Computer Science, Hanyang University